

Freeze-out of light nuclei at low and intermediate energies

[arXiv:2605.11275]

Oleh Savchuk

APS DNP, October 11-14th



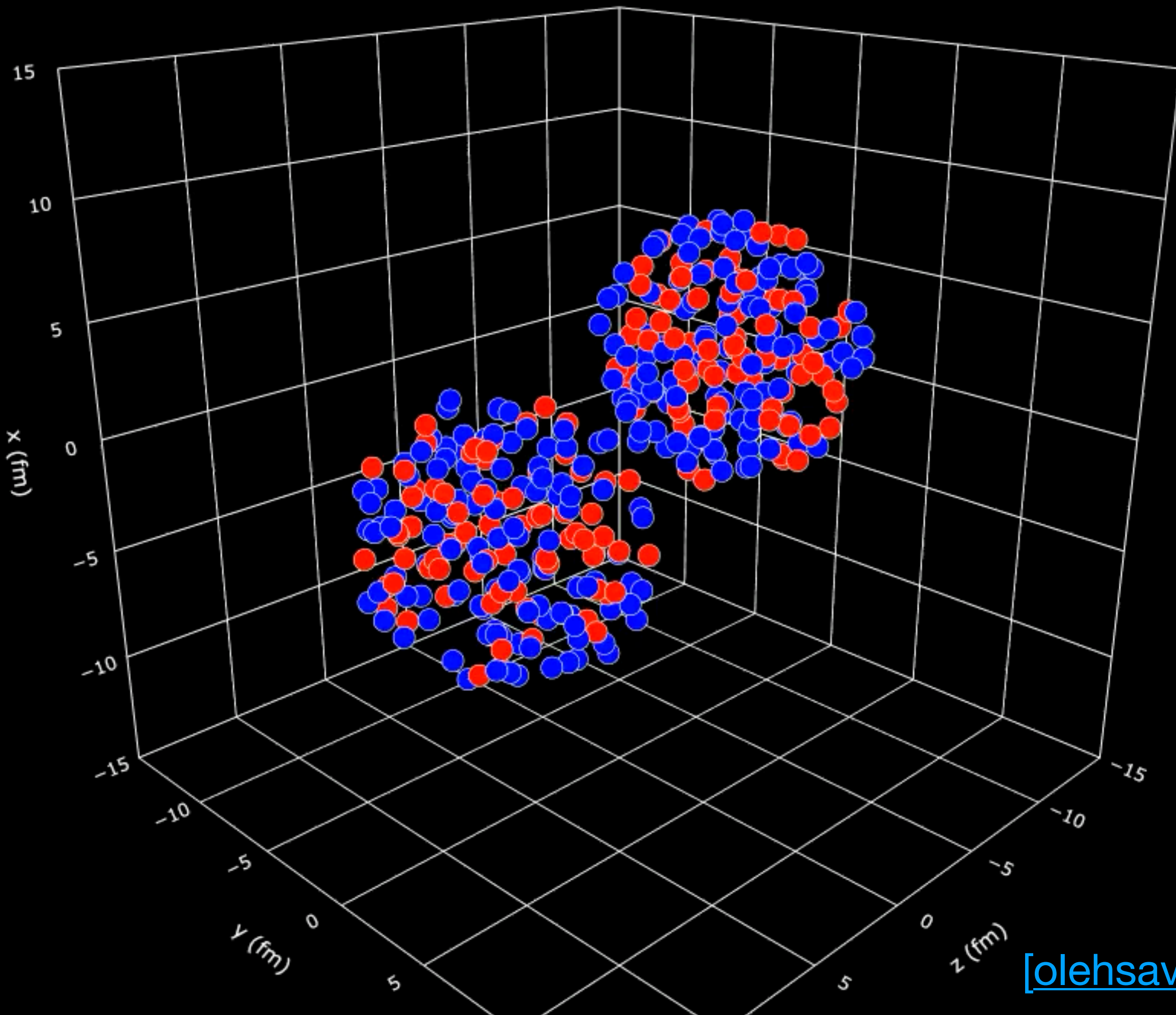
**Facility for
Rare Isotope Beams**
at Michigan State University



Light nuclei formation in collisions

2

Heavy-Ion Collision sketch:



particle
● Proton
● Neutron

Nuclei are smashed together

System undergoes compression and pressure build-up

During the expansion light clusters are formed

We developed an approach to predict light nuclei production based on **time dependent** local thermal equilibrium

Software package available at:

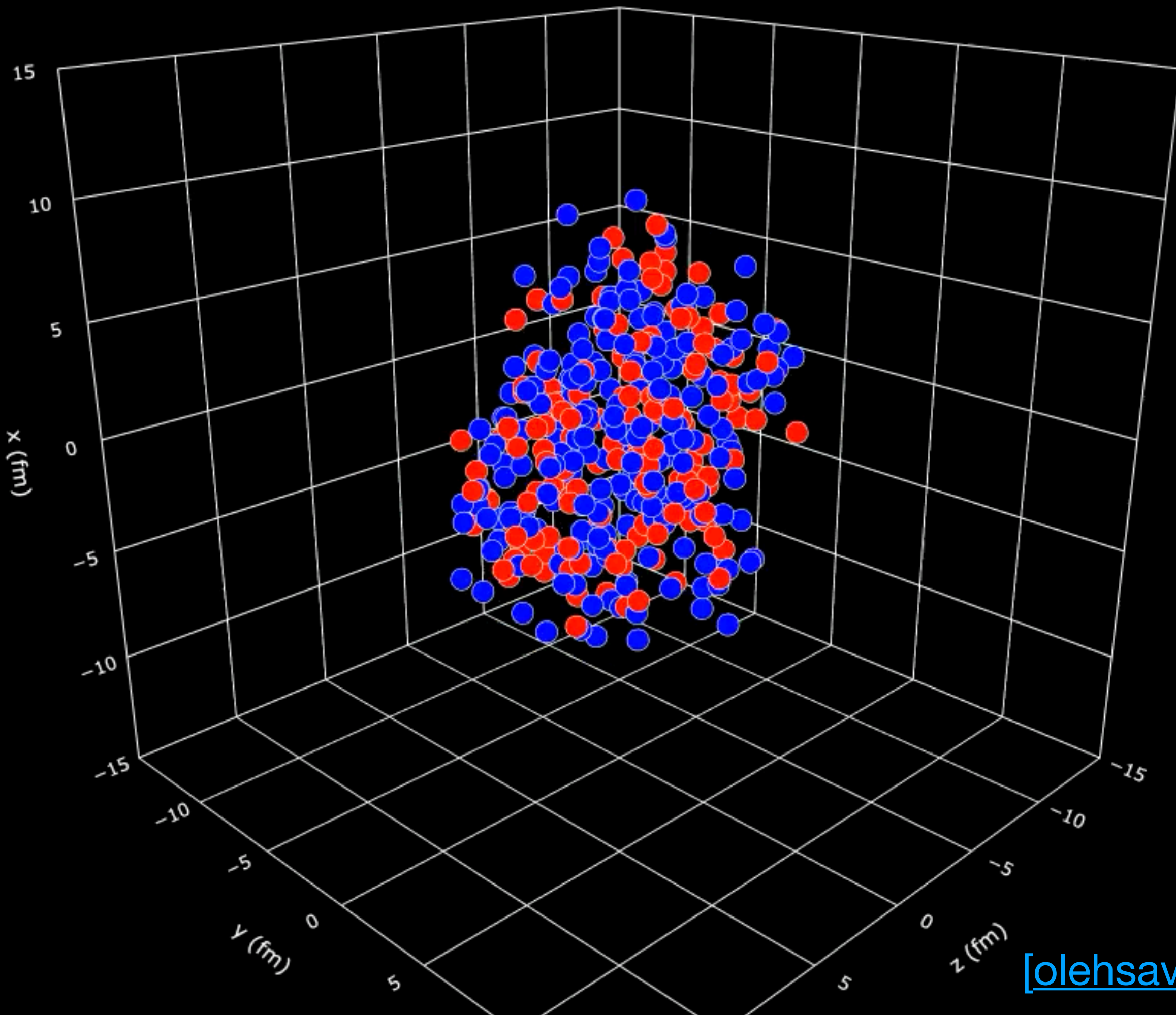
[\[olehsavchuk.github.io/plots\]](https://olehsavchuk.github.io/plots) [\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)



Light nuclei formation in collisions

3

Heavy-Ion Collision sketch:



particle
● Proton
● Neutron

Nuclei are smashed together

System undergoes compression and pressure build-up

During the expansion light clusters are formed

We developed an approach to predict light nuclei production based on **time dependent** local thermal equilibrium

Software package available at:

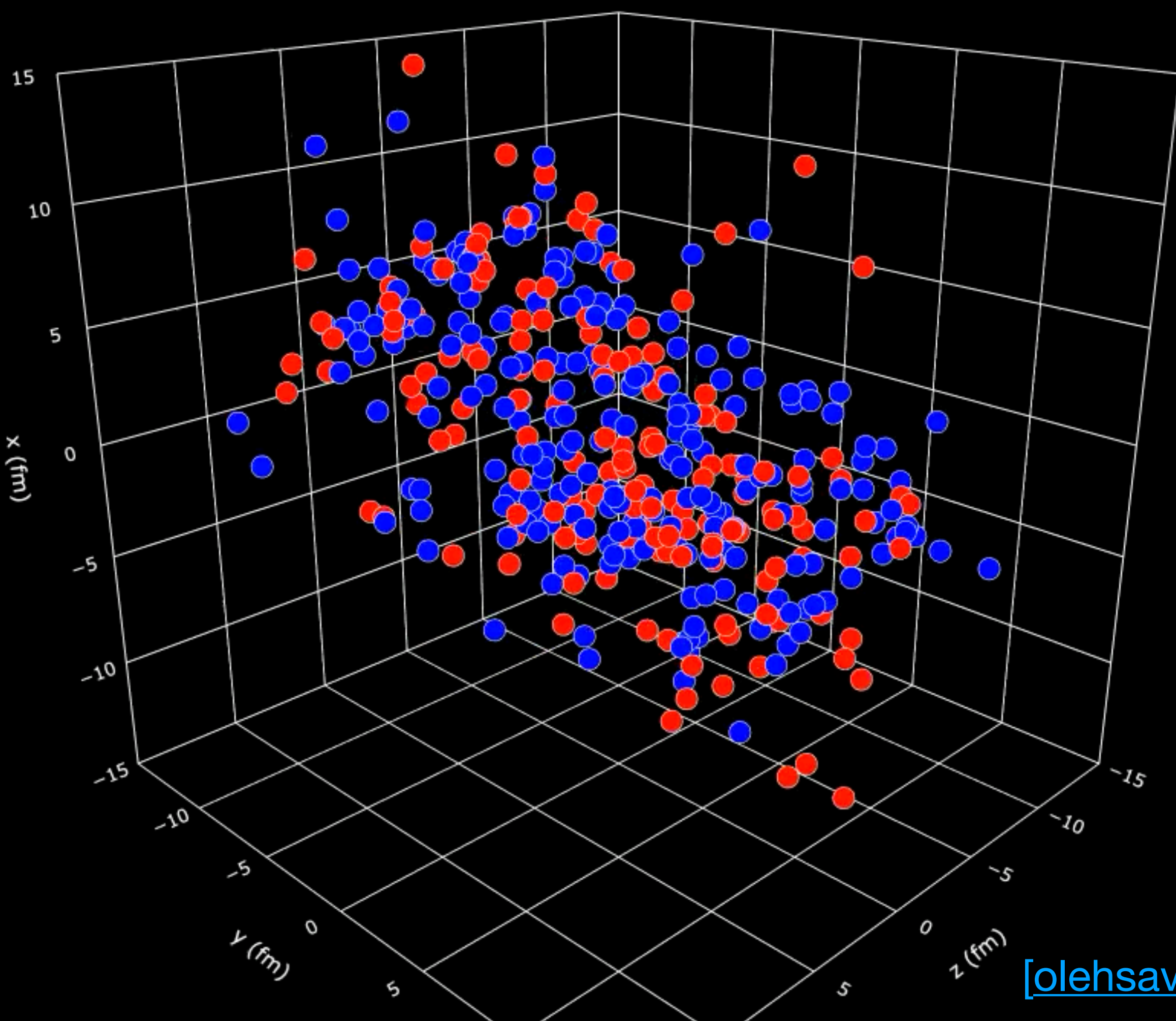
[\[olehsavchuk.github.io/plots\]](https://olehsavchuk.github.io/plots) [\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)



Light nuclei formation in collisions

4

Heavy-Ion Collision sketch:



particle
● Proton
● Neutron
● ID: -3.0

Nuclei are smashed together

System undergoes compression and pressure build-up

During the expansion light clusters are formed

We developed an approach to predict light nuclei production based on **time dependent** local thermal equilibrium

Software package available at:

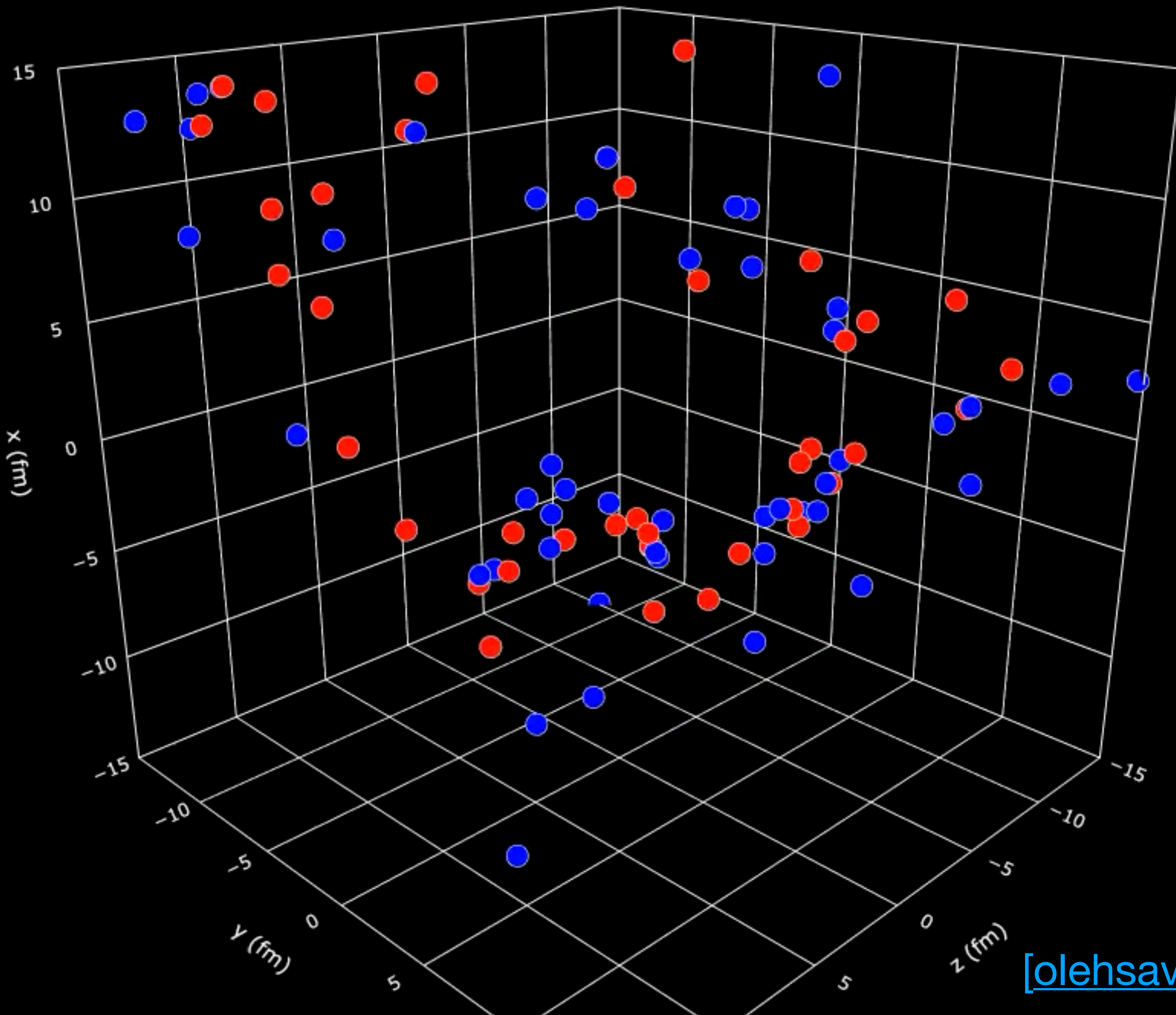
[\[olehsavchuk.github.io/plots\]](https://olehsavchuk.github.io/plots) [\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)



Light nuclei formation in collisions

5

Heavy-Ion Collision sketch:



particle
● Proton
● Neutron

Nuclei are smashed together

System undergoes compression and pressure build-up

During the expansion light clusters are formed

We developed an approach to predict light nuclei production based on **time dependent** local thermal equilibrium

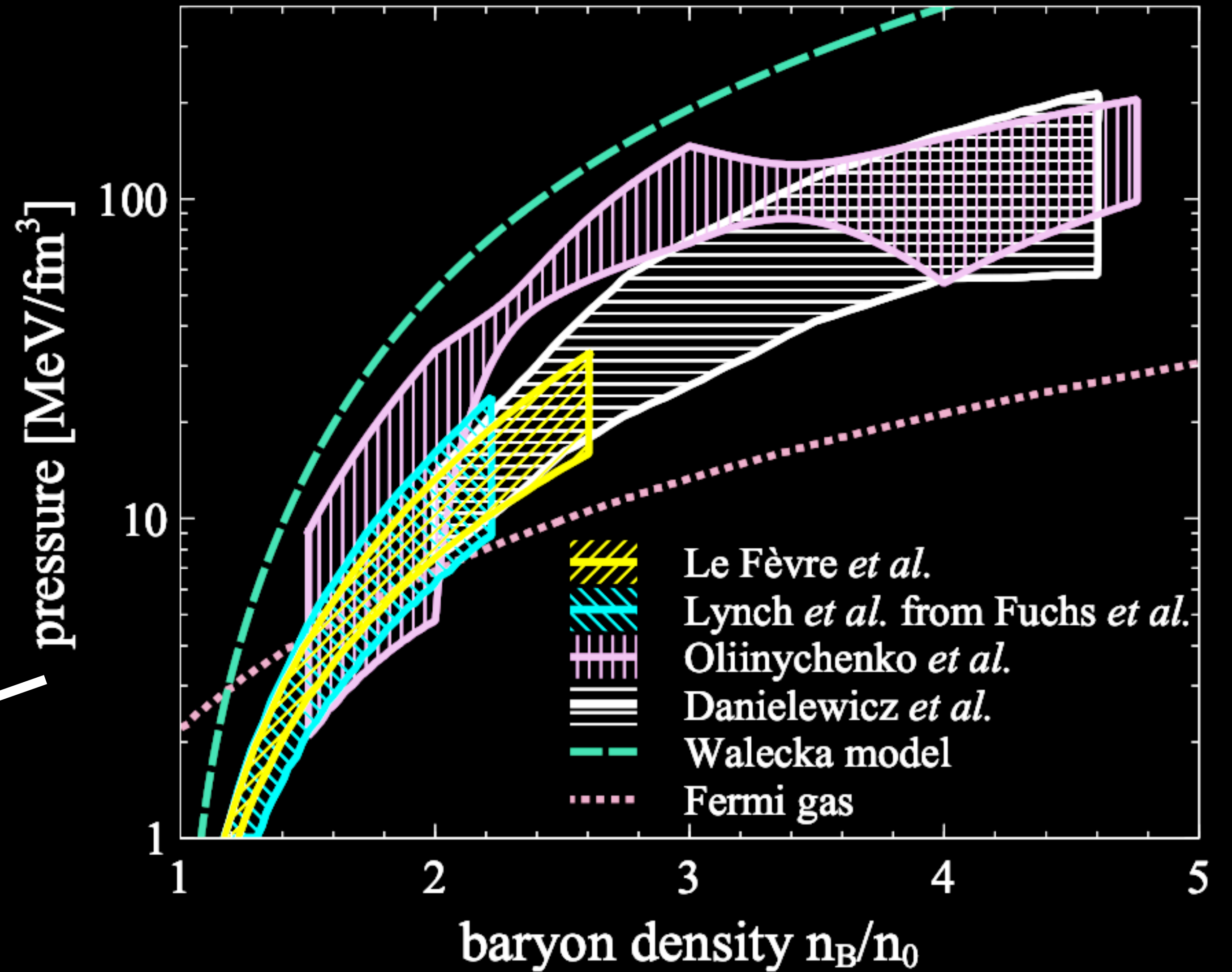
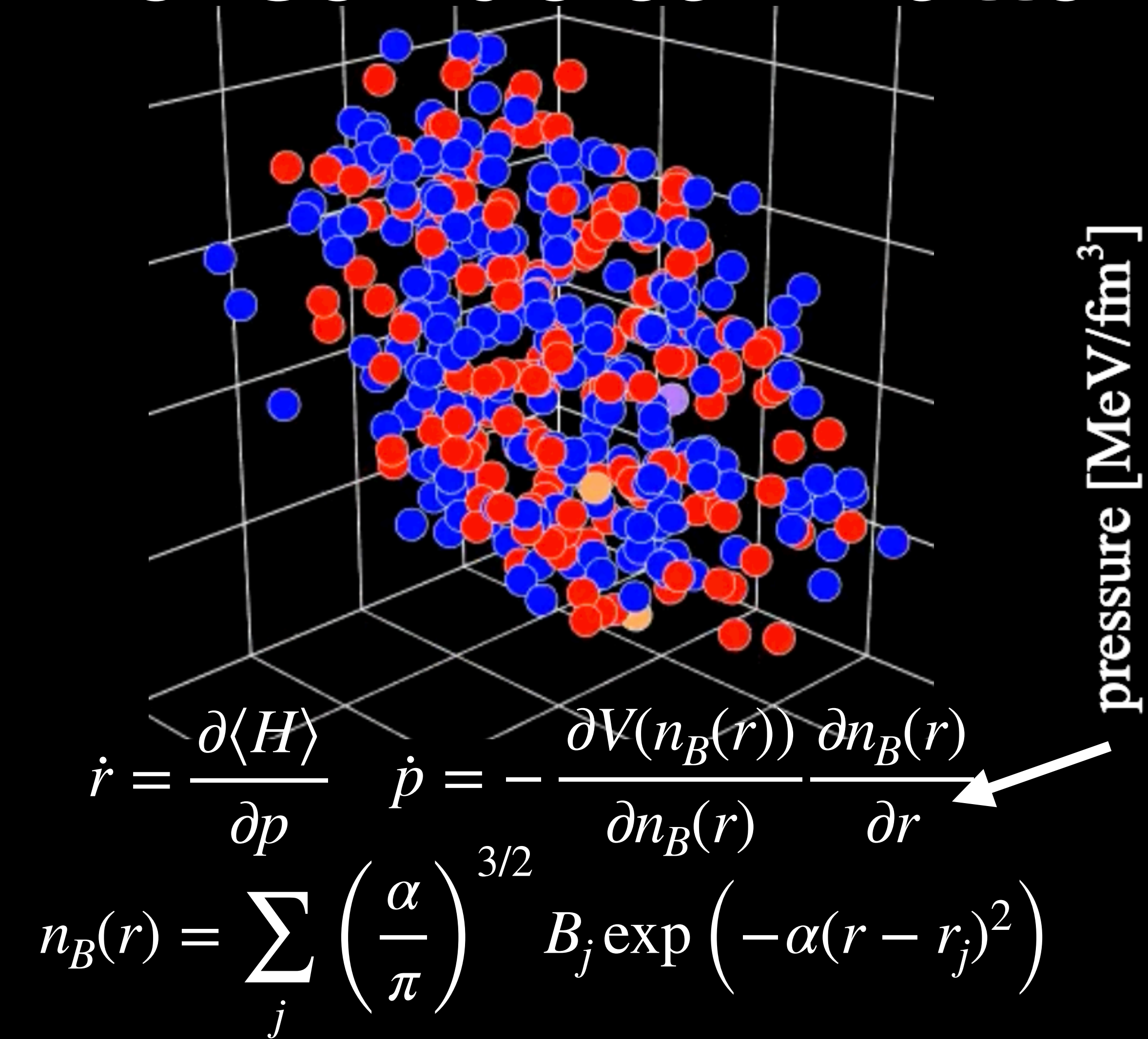
Software package available at:

[\[olehsavchuk.github.io/plots\]](https://olehsavchuk.github.io/plots) [\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)



Dense nuclear matter

[Prog.Part.Nucl.Phys. 134 (2024) 104080]

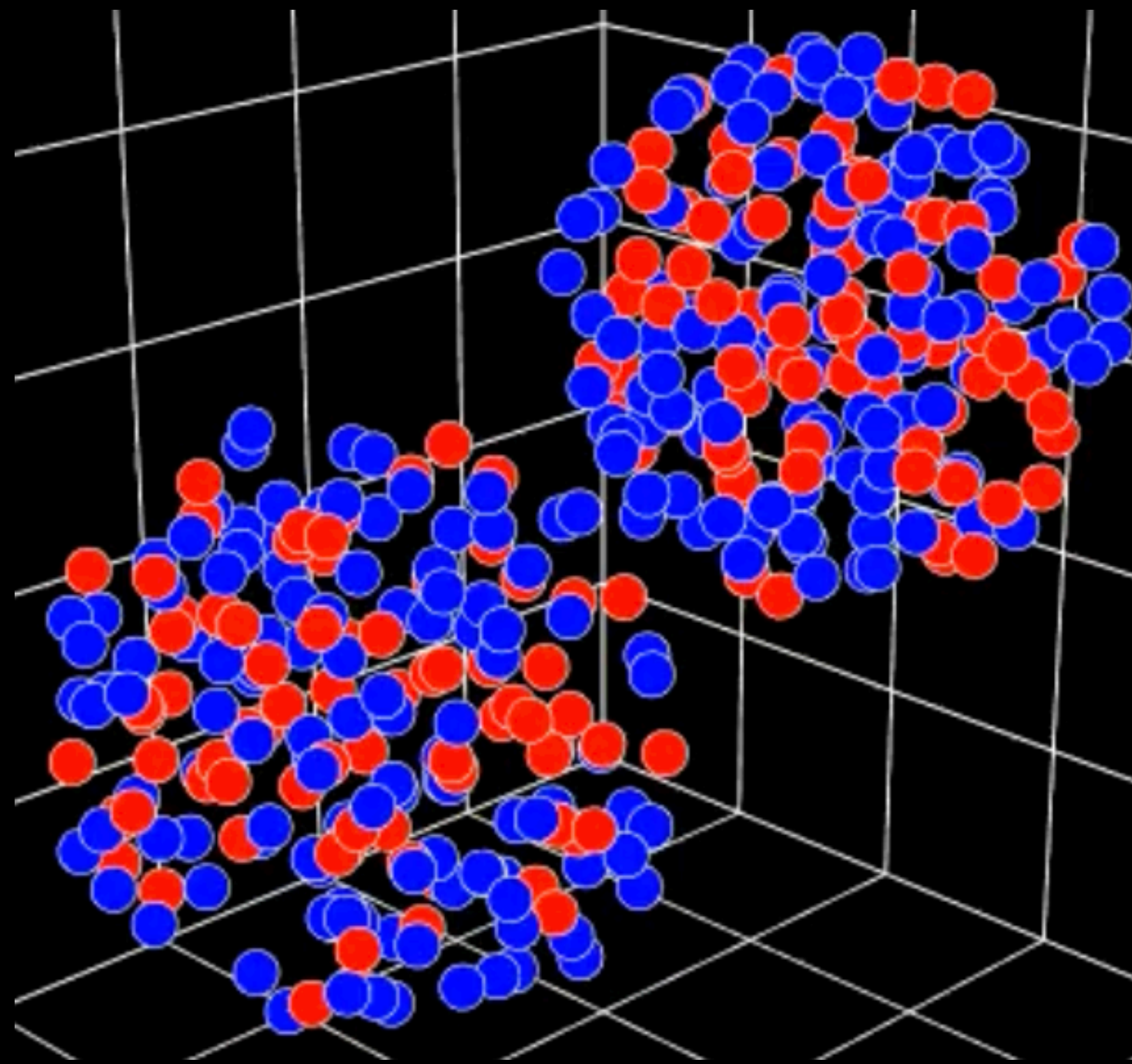


Propagate wave-packets in density dependent potentials between stochastic scattering

Light clusters affect most observables

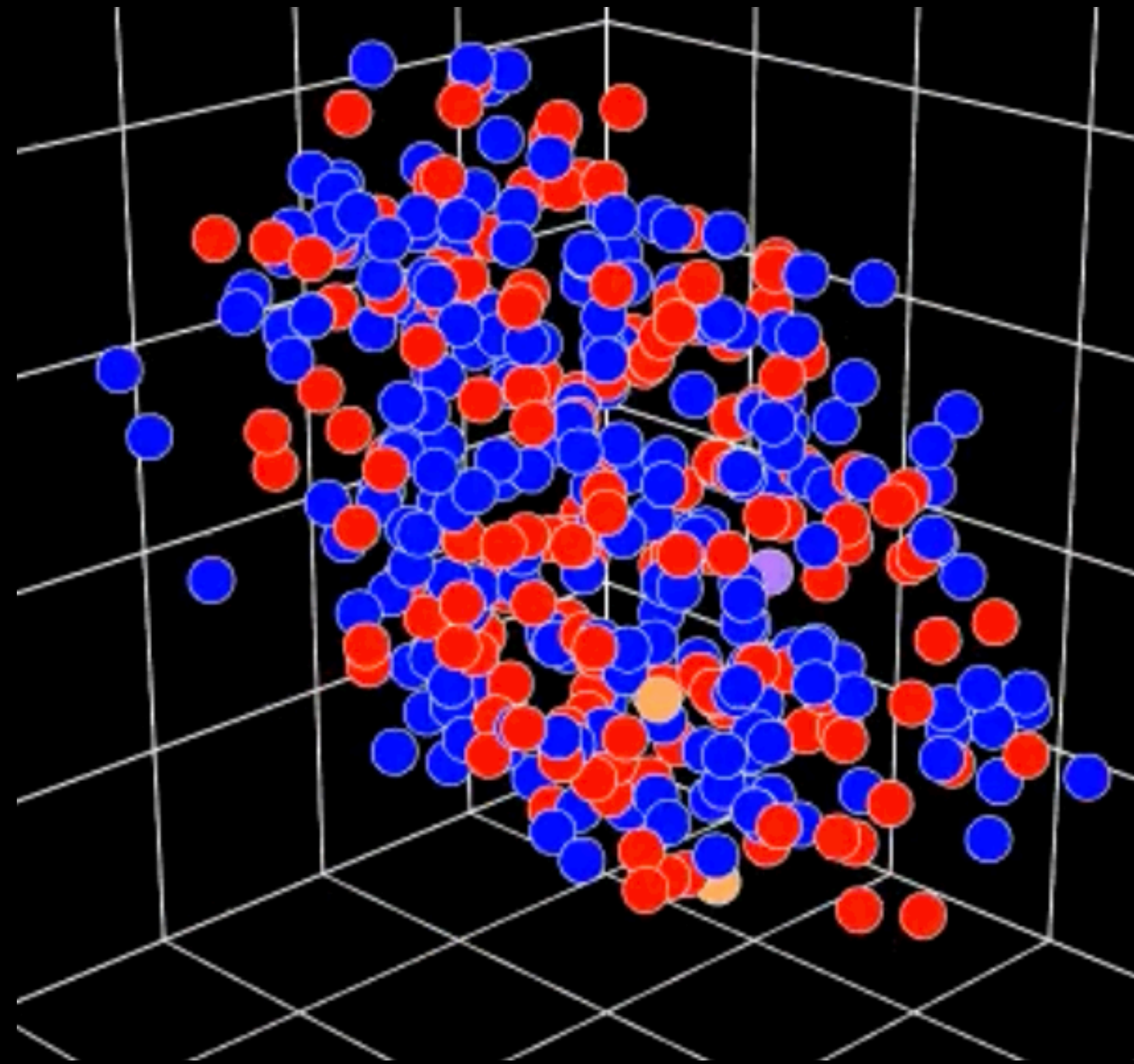
7

Initial stage - cold
nuclear matter properties



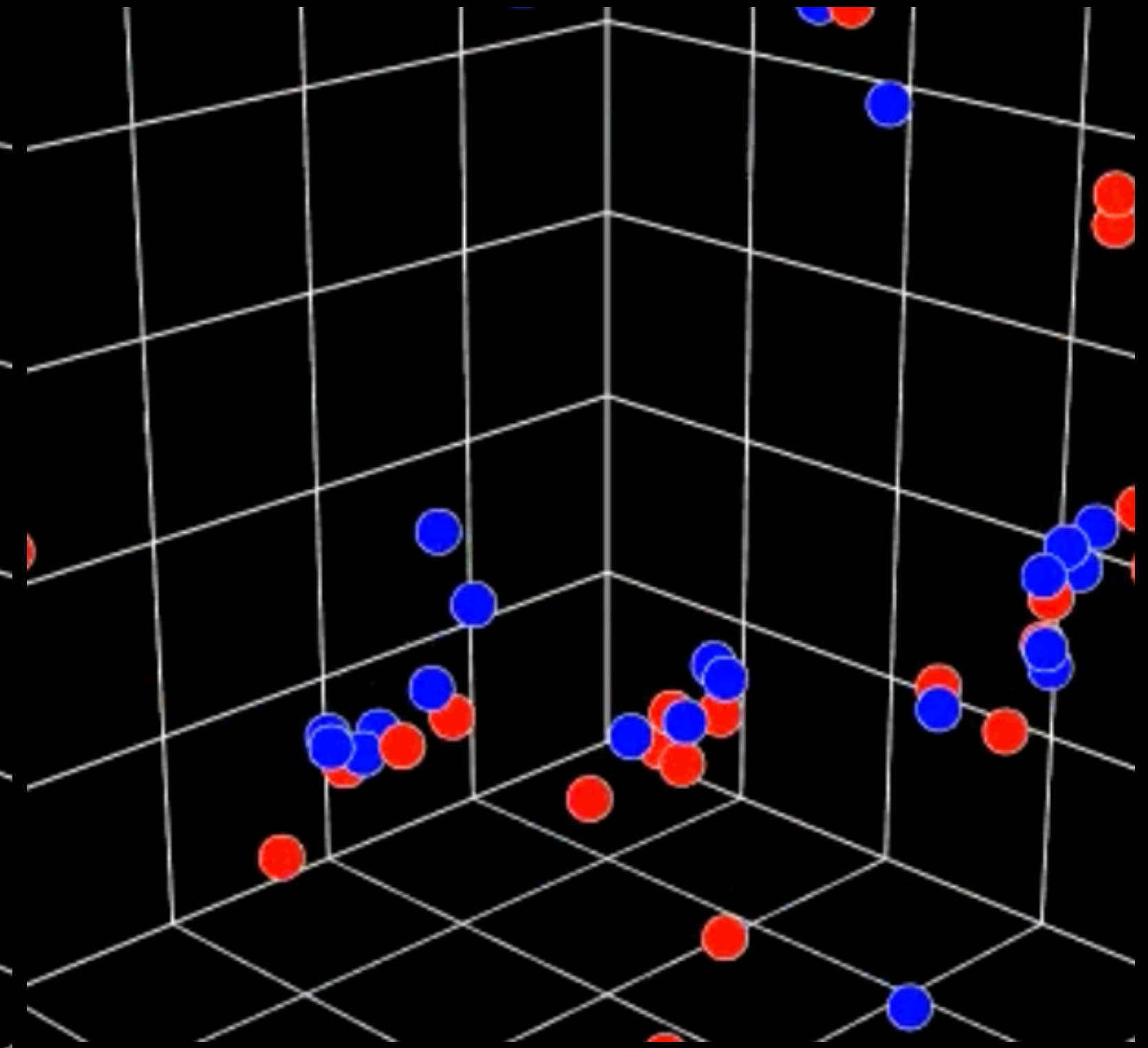
Semi understood
 $n \approx n_0$

Hot and dense stage
“explosion”



Pressure build-up
 $n > 2n_0$

Freezing out of
interactions and detection



Gas of light fragments
 $n_B < n_0$

Despite being at the lowest density, late stages are notoriously hard to model due to sheer amount of reactions involving light clusters

[\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)

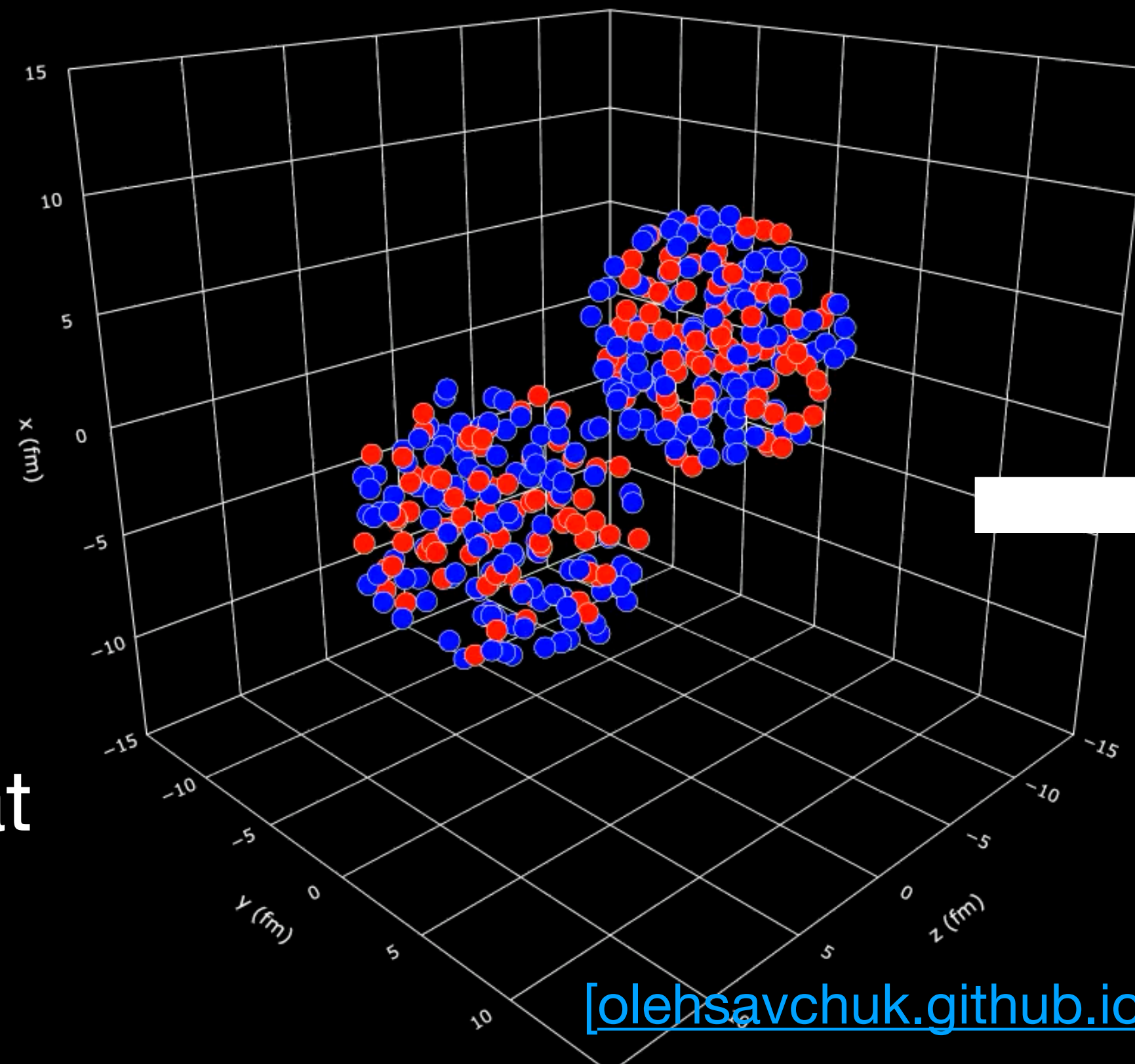
Local thermal equilibrium

8

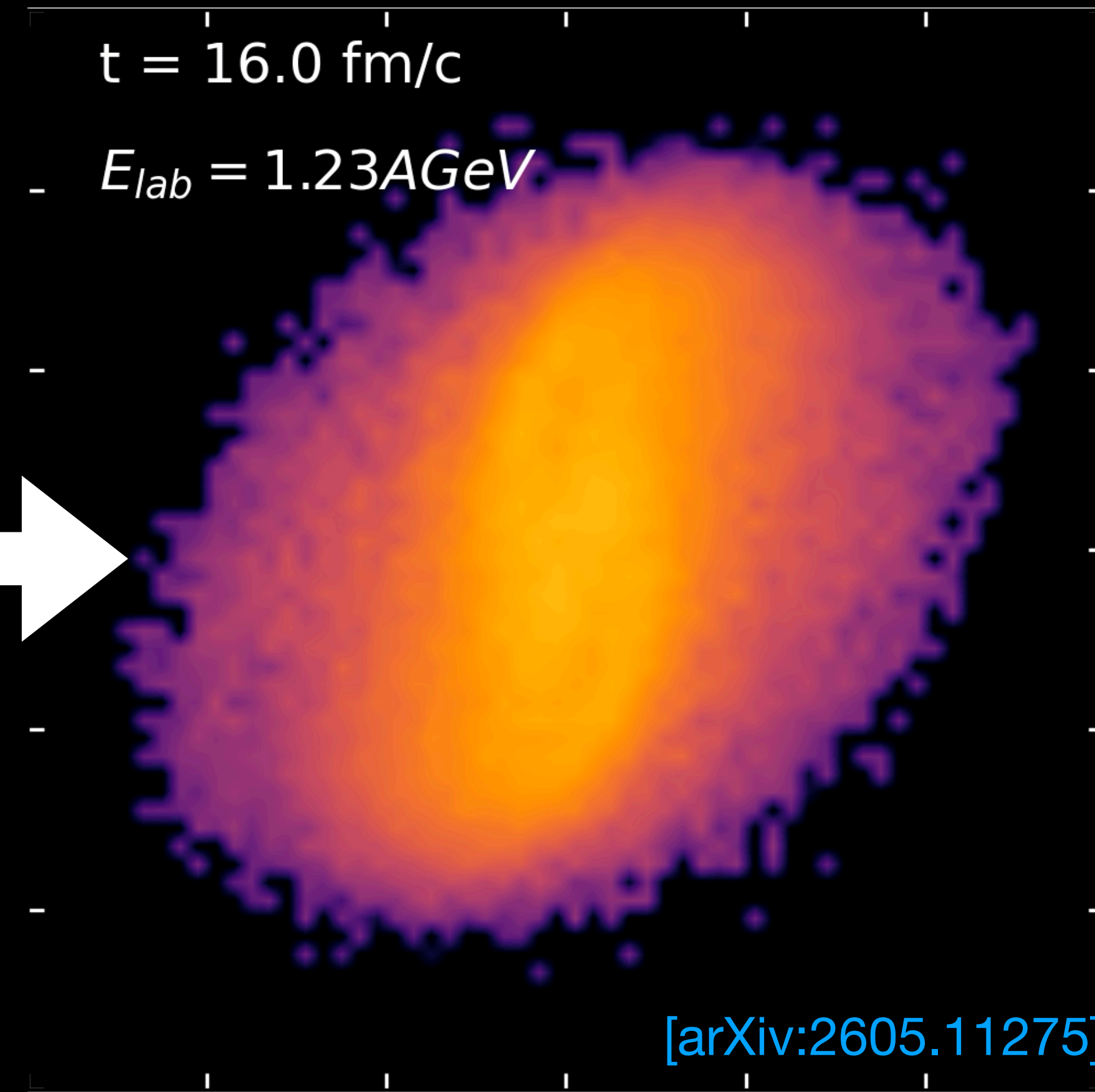
Thermodynamics of a large ensemble of similarly prepared heavy-ion collisions

Clusters are defined by the temperature and density at emission

Ensemble of collisions



Temperature



Freeze-out occurs at
 $n_B = n_0/8$

[\[olehsavchuk.github.io/plots\]](https://github.com/olehsavchuk/plots)

[\[arXiv:2605.11275\]](https://arxiv.org/abs/2605.11275)

Statistical model of clusters

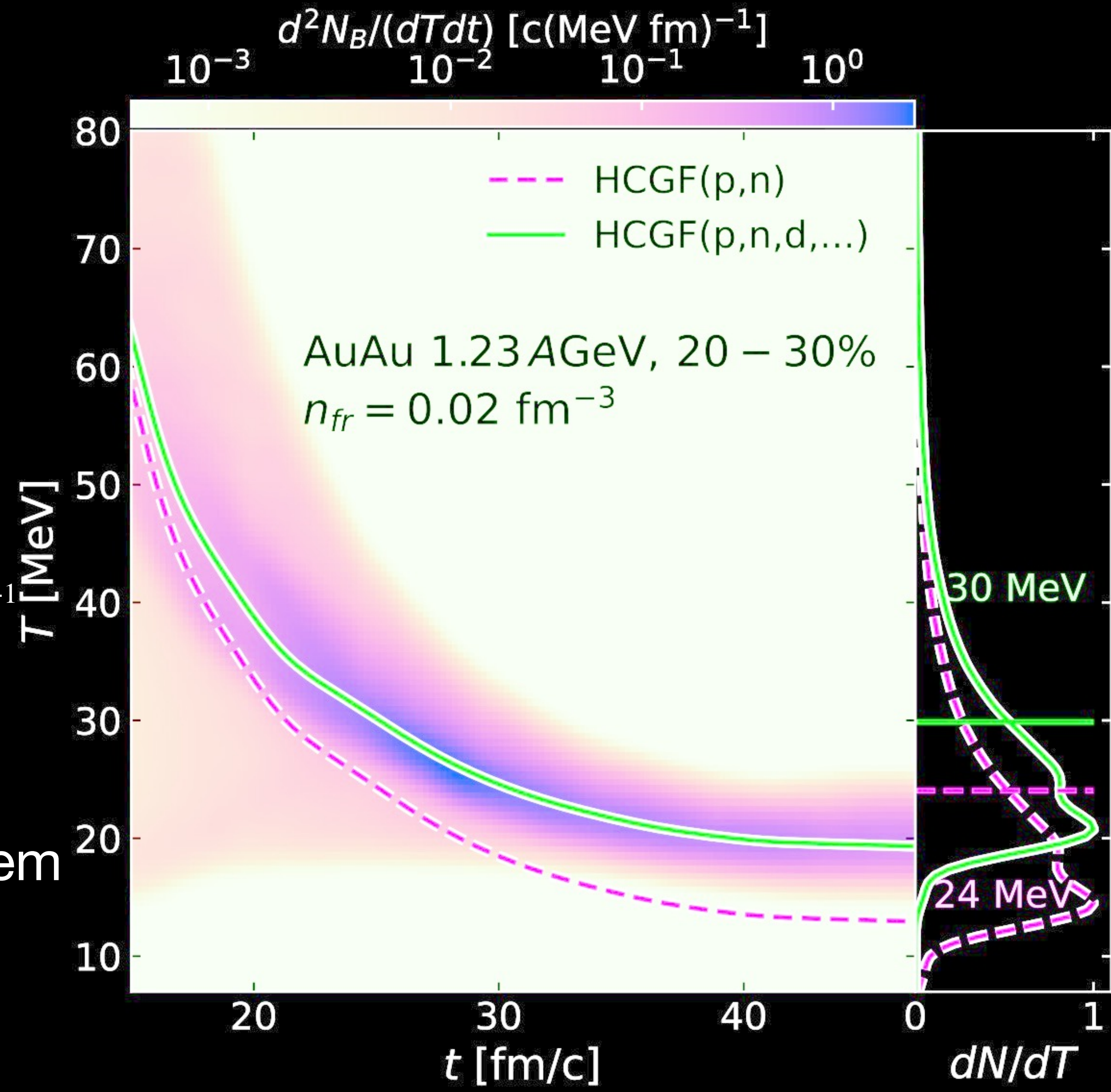
We include nucleons p, n , light fragments $d, t, {}^3\text{He}, {}^4\text{He}$

excited states ${}^4\text{He}^*, \dots$ and unstable isotopes e.g. ${}^5\text{Li}$

Phase space densities defined by $\vec{v}, T, \mu$ matched to energy and baryon density of the simulation:

$$f_i(x, p) = \frac{d^6 N_i}{d^3 x d^3 p} = \frac{g_i}{(2\pi)^3} \left[\exp \left(\frac{\gamma(\sqrt{m_i^2 + p^2} - \vec{p} \cdot \vec{v}) - \mu_i}{T} \right) + \eta_i \right]^{-1}$$

Equilibrium distr. + boost to laboratory system



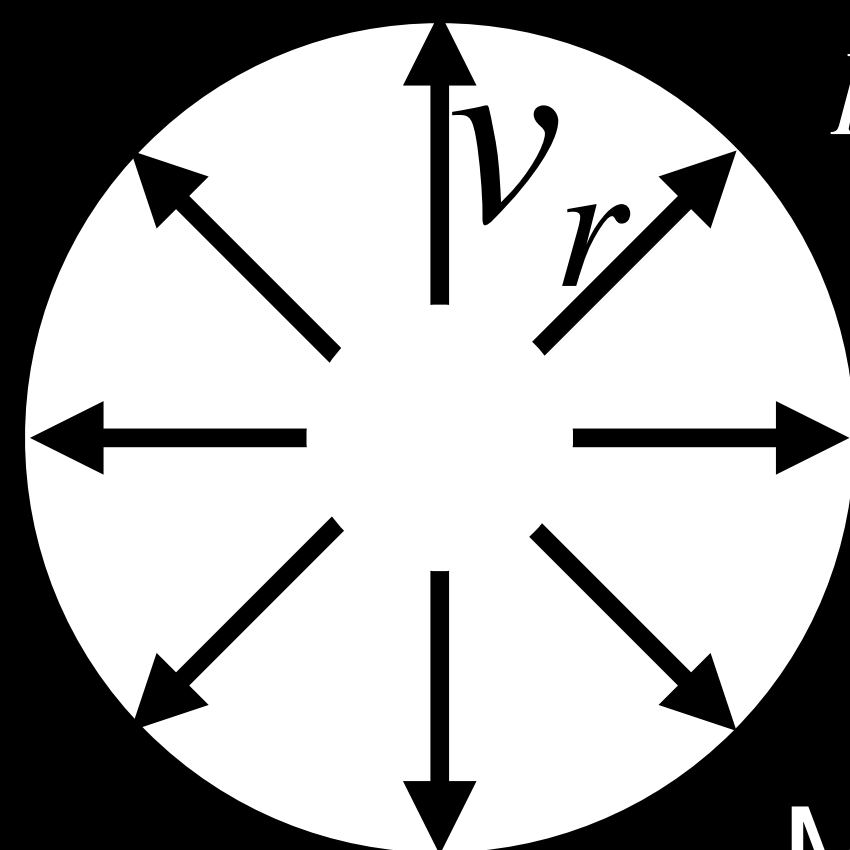
Transverse spectra

Clustering changes shape of proton spectra

Tentative signal of temperature increase

Light fragments and free protons add up to the proton spectra before clustering

Spectra includes both contributions from temperature and radial expansion



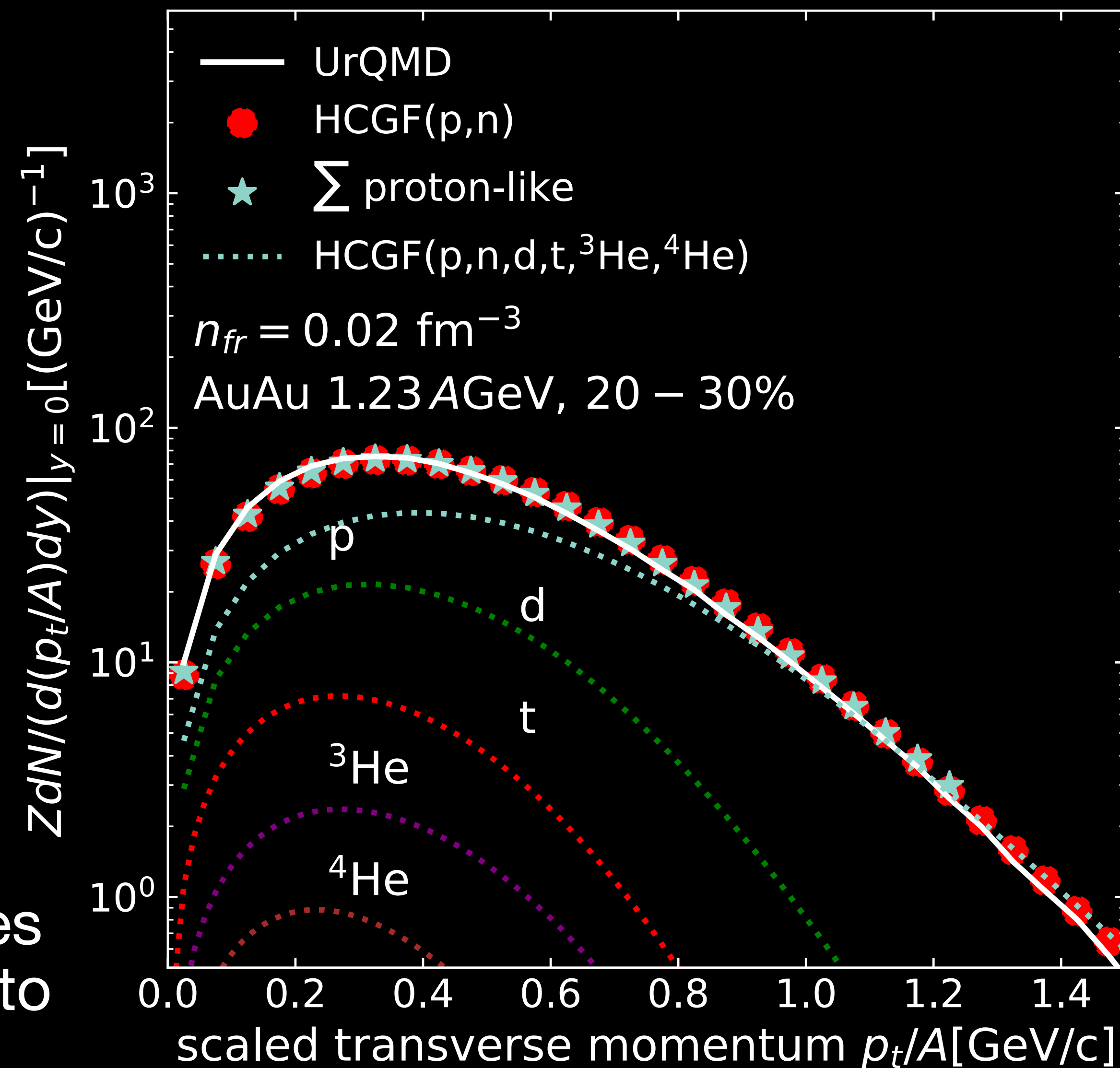
$$p \approx p_{coll} + p_{th}$$

$$p_{coll} \approx mv_r$$

$$p_{th} \sim \sqrt{mT}$$

Heavier particles
more sensitive to
the flow

Mass scaling indicates same source for different particle species



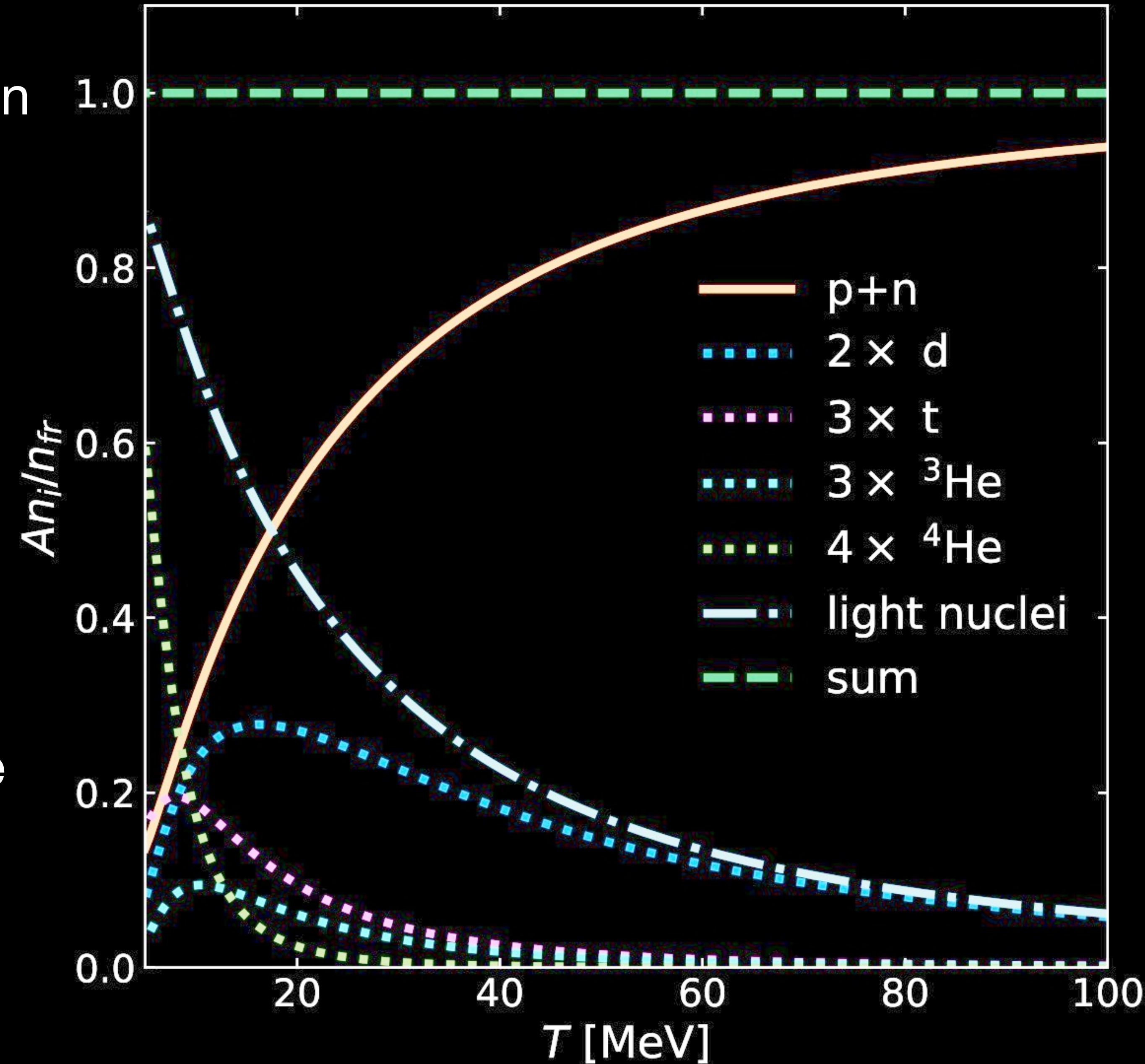
Cluster fractions

11

In non-interacting system yields are driven by the entropy

At high temperature clustering is suppressed

At the low temperature clusters dominate the yields

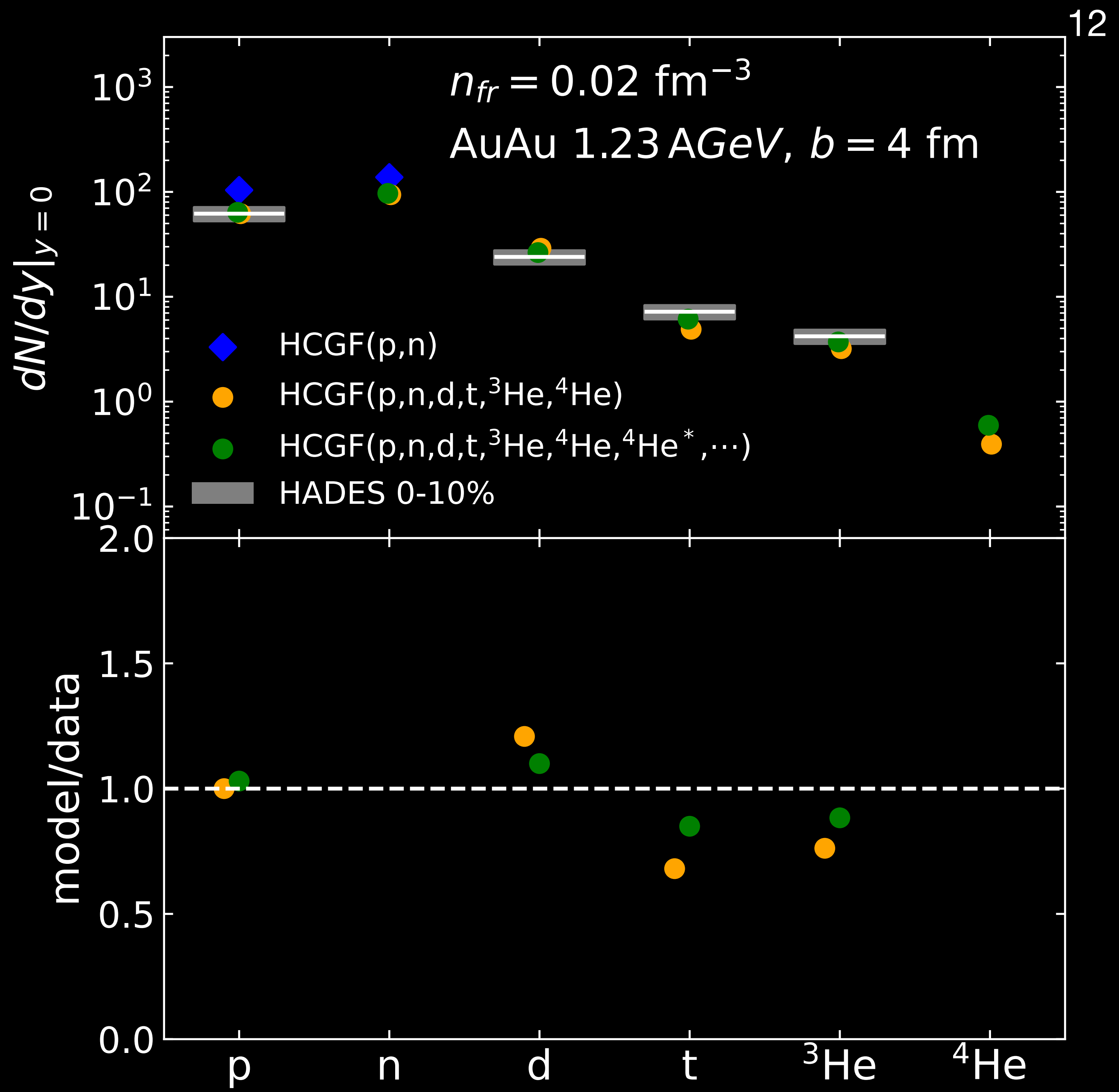


Particle yields

We calculate measured stable isotope yields by including decays of unstable ones

This leads to correct description of the data

We extract yields directly from the simulation, thus establishing a link between observables and microscopic transport model



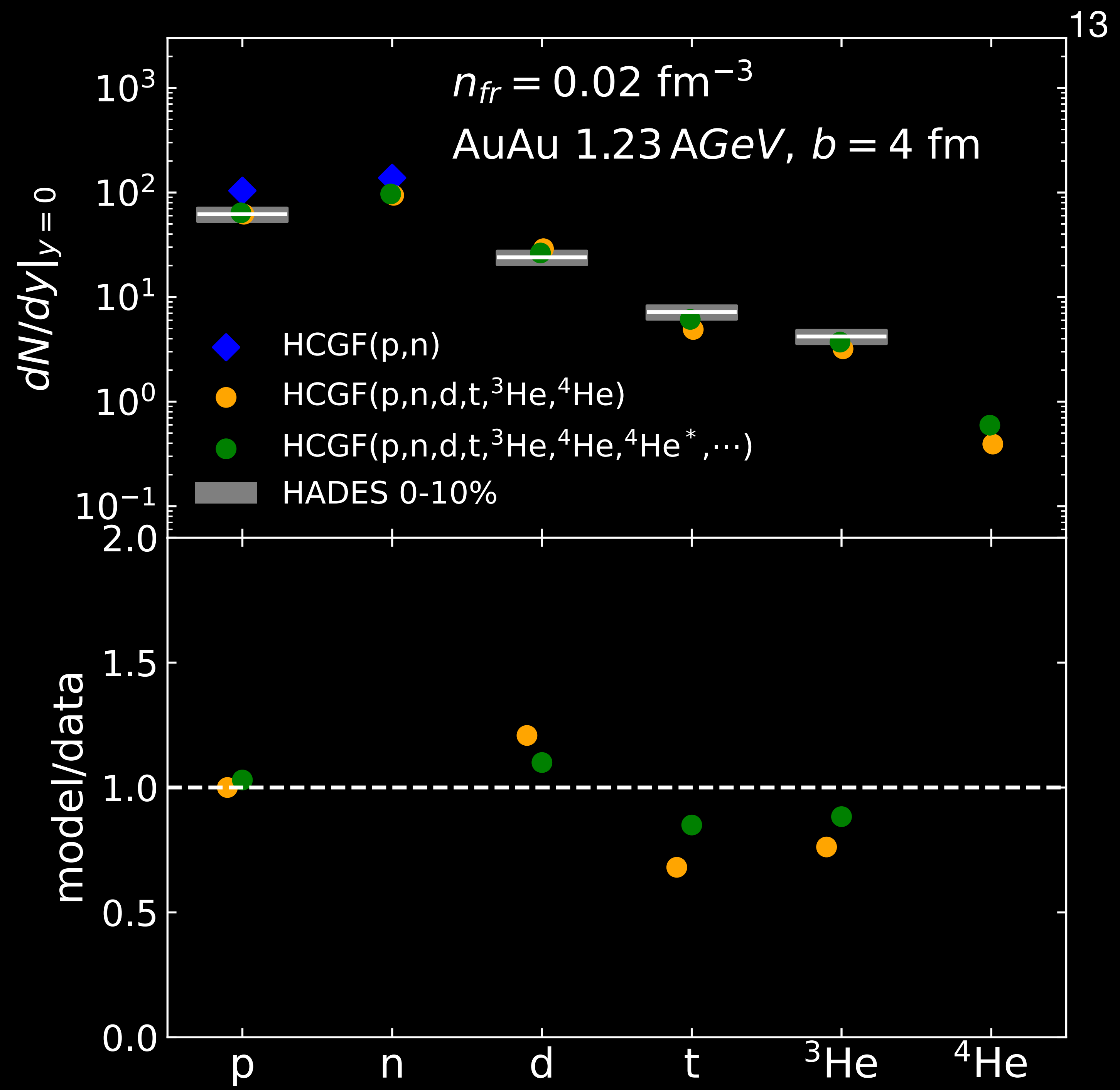
Conclusions

A novel approach to cluster formation is suggested

Spectra and multiplicities of light fragments were calculated directly from transport simulation

Heavier fragments and excited states were included in the simulation

Should allow us to constrain the equation of state better



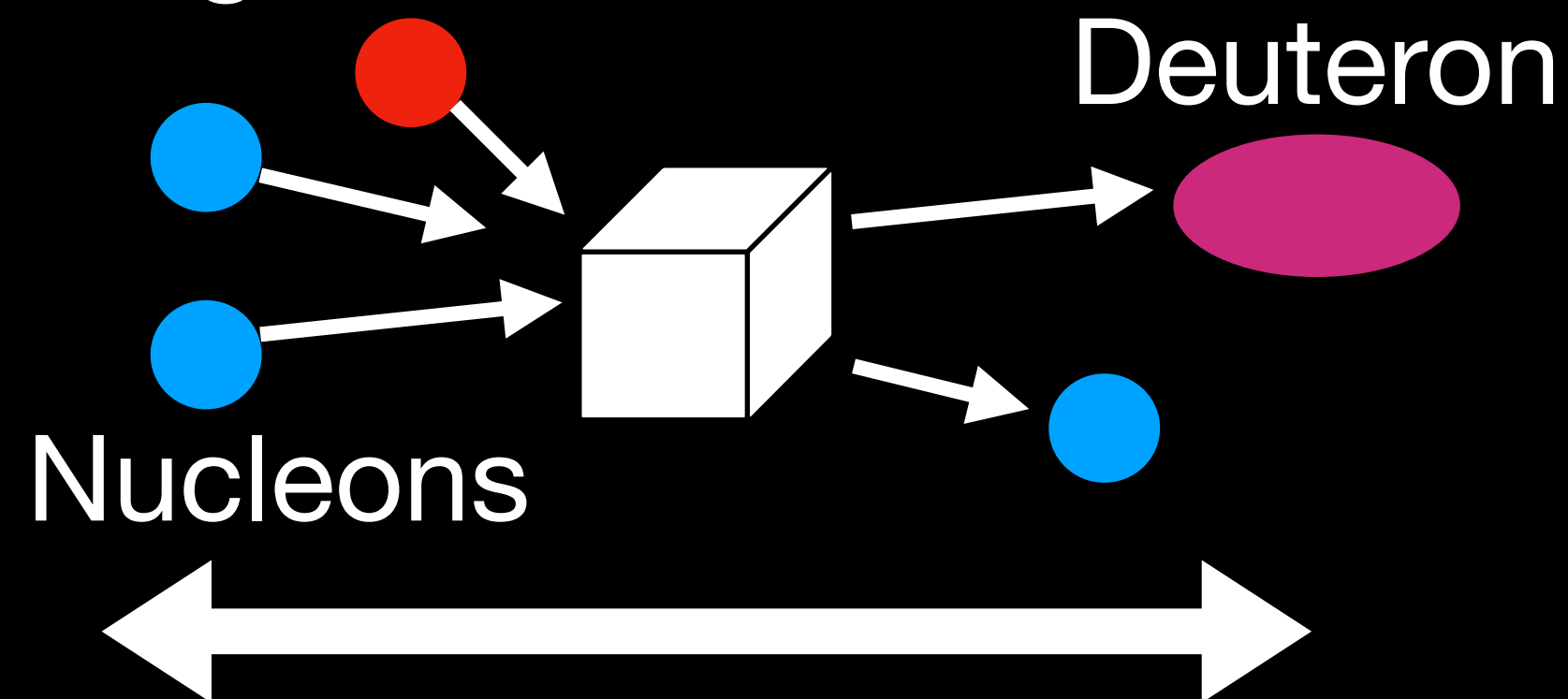
Clustering in transport models

14

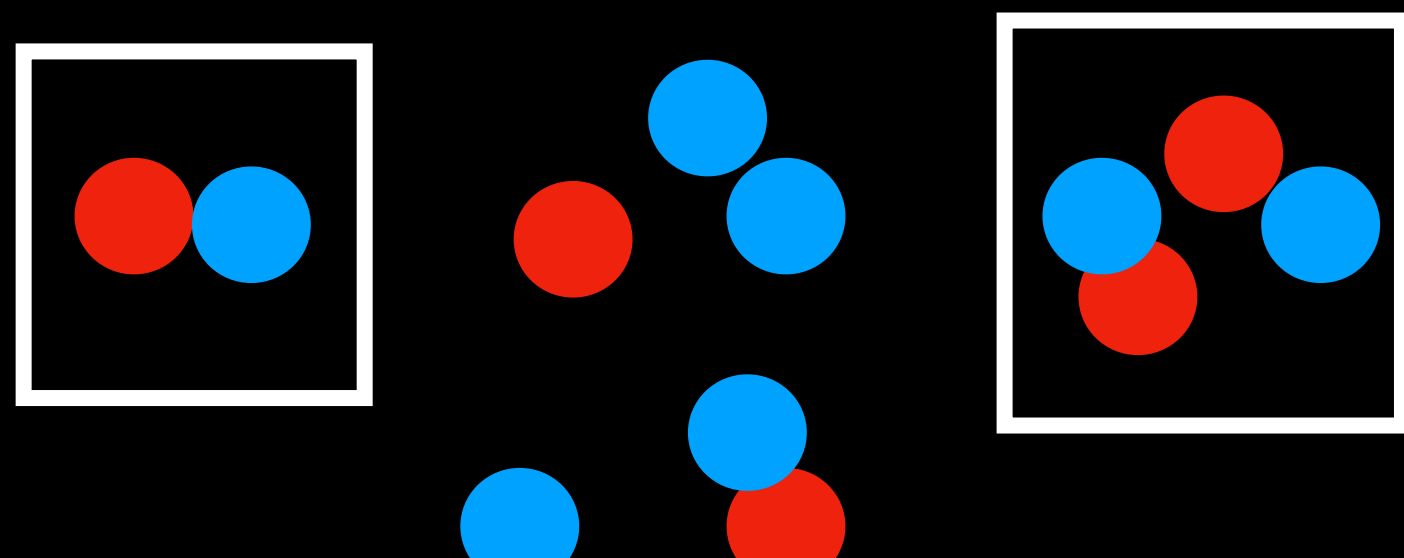
In-medium clustering:

Snowballs in hell

1. Clusters as separate particle species that are propagated alongside nucleons

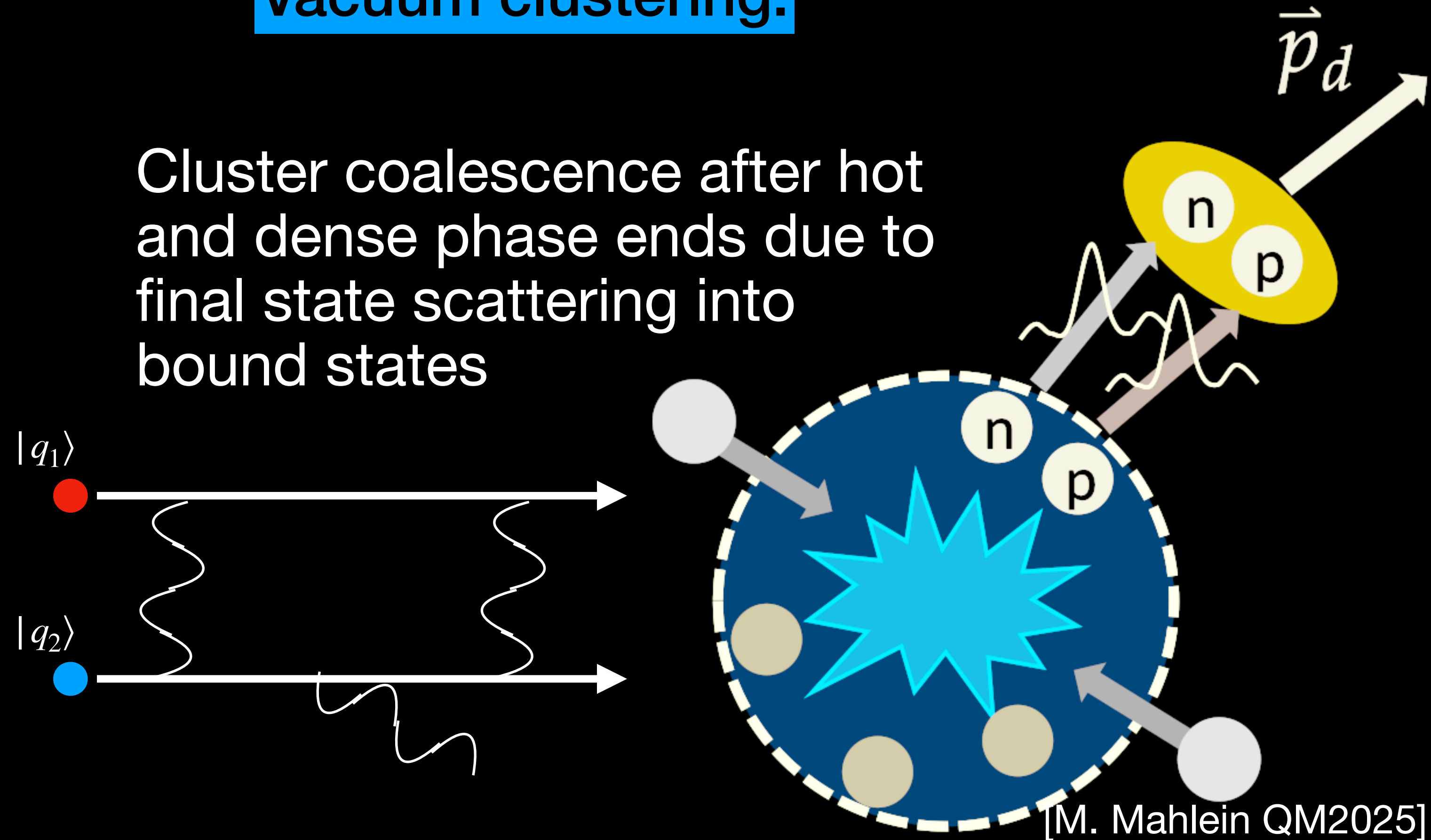


2. Interacting nucleons cluster naturally - cluster identification algorithms are required



Vacuum clustering:

Cluster coalescence after hot and dense phase ends due to final state scattering into bound states



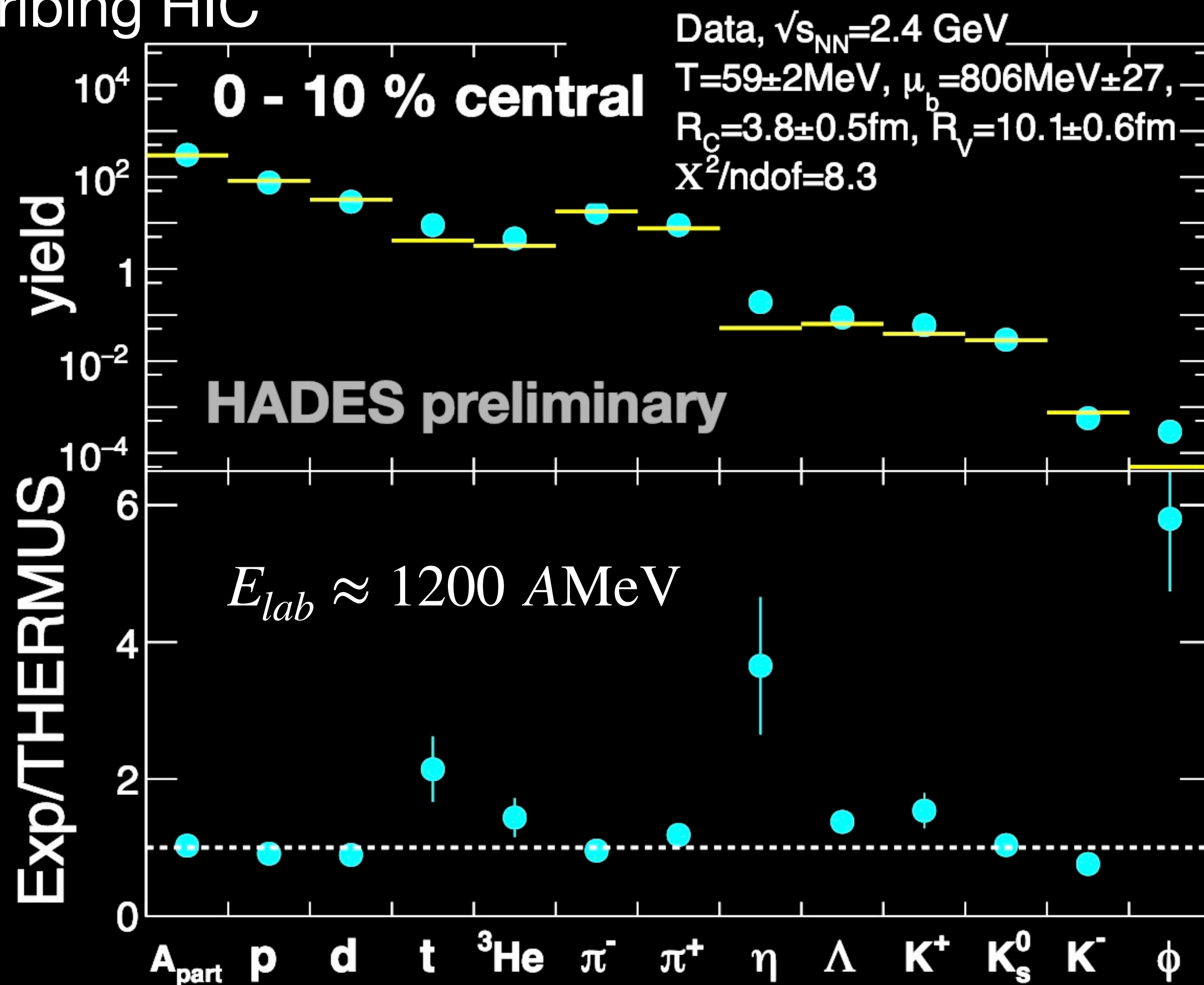
Usually very simplified but still have a lot of parameters that require fine tuning

Thermal model success

Equilibrium is very successful in describing HIC

In particular yields of particles can be described assuming a single source with T, μ_B, μ_Q, μ_S

These thermodynamic parameters are usually fitted to the data, but one can extract them from transport linking them to properties of the dense matter

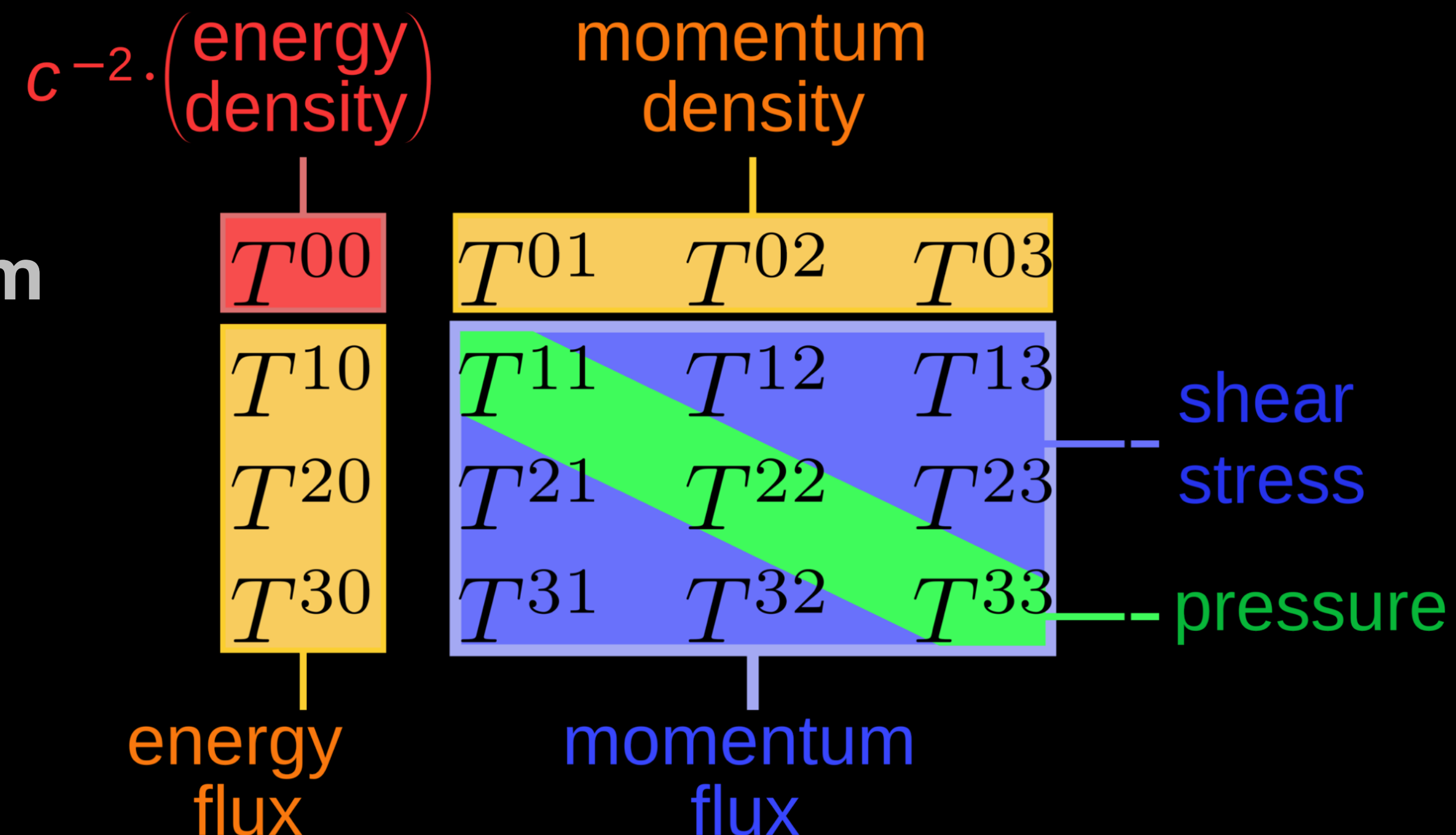


Equilibrium stress energy tensor

The stress-energy tensor is a symmetric tensor which describes the spacetime density and flux of energy and momentum

$$T^{\mu\nu} = \left\langle \sum_i B_i \frac{p^\nu p^\mu}{p^0} \right\rangle / V + T^{\mu\nu}_{fields}$$

$$T^{\mu\nu}_{EM} = F^{\mu\alpha} F^\nu_\alpha - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}$$



In full equilibrium stress energy tensor has to be diagonal in the local rest frame (defined by baryon current):

$$T^{\mu\nu}_{\text{LRF}} = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

But the tensor extracted from transport will not look like that

Because equilibrium is not ideal and system still relaxes towards it

Stress-energy tensor out of equilibrium

17

$$T^{\mu\nu} = \epsilon u^\mu u^\nu - (P^{eq} + \Pi) \Delta^{\mu\nu} + \pi^{\mu\nu} + q^\mu u^\nu + q^\nu u^\mu$$

$$\Delta^{\mu\nu} = g^{\mu\nu} - u^\mu u^\nu$$

$$\epsilon = u_\mu u_\nu T^{\mu\nu} \quad \text{Energy density}$$

$$q^\mu = \Delta^\mu_\alpha T^{\alpha\beta} u_\beta \quad \text{Heat flux, defines difference in charge and energy transfer}$$

$$P^{eq} + \Pi = -\frac{1}{3} \Delta_{\mu\nu} T^{\mu\nu} \quad \text{Bulk pressure, connected to bulk viscosity and defines how pressure changes when volume changes fast } \Pi = -\zeta \partial_\mu u^\mu,$$

$$\pi^{\mu\nu} = \left(\Delta^\mu_\alpha \Delta^\nu_\beta - \frac{1}{3} \Delta^{\mu\nu} \Delta_{\alpha\beta} \right) T^{\alpha\beta} \quad \text{Shear viscosity, Navier-Stokes term}$$

Distribution function has to match $T^{\mu\nu}$:

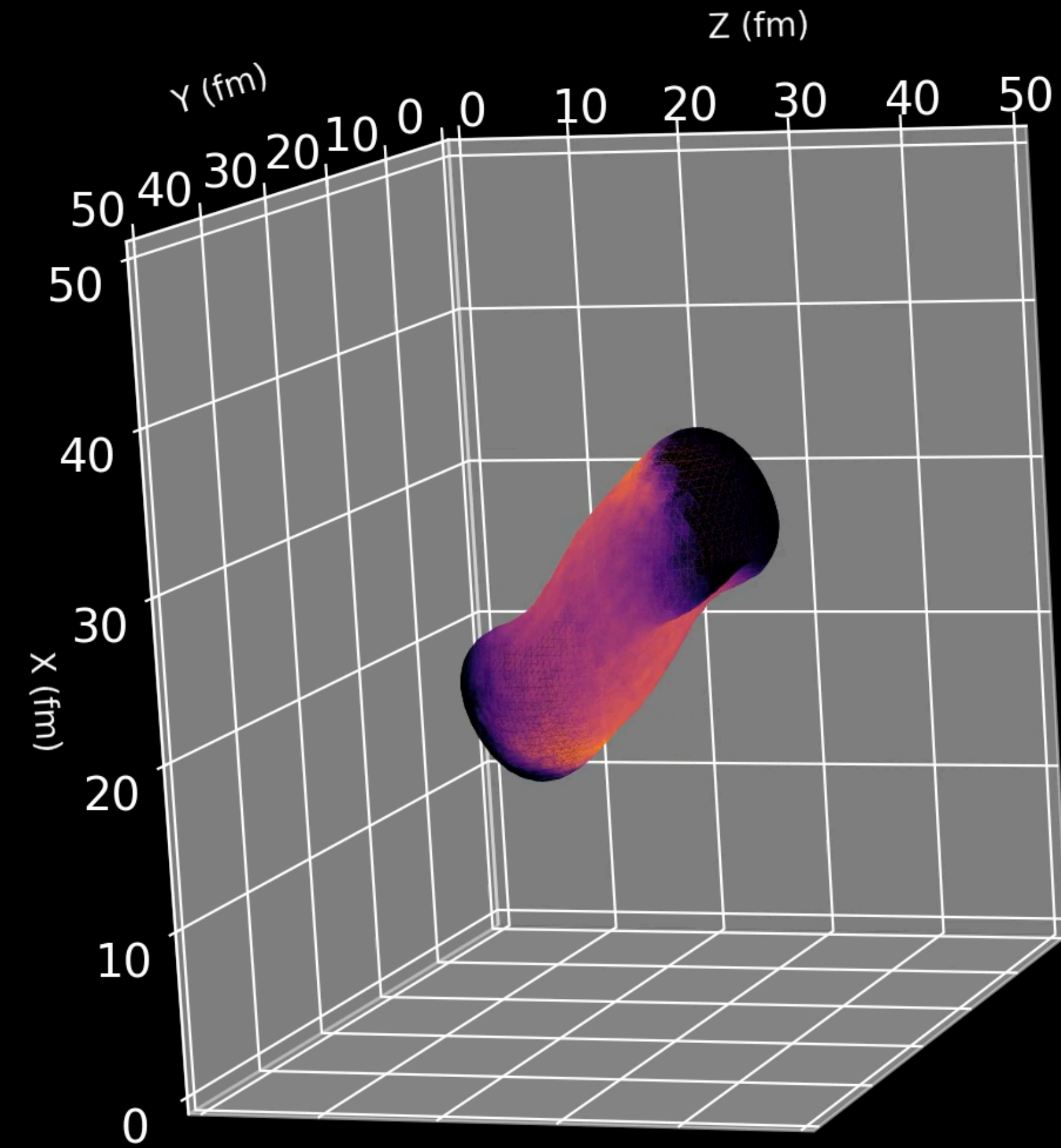
$$T^{\mu\nu} = \int \frac{d^3 p}{p_0} f(t, x, p) p^\mu p^\nu$$

Freeze-out hyper surface

18

Temperature Distribution Comparison | Time = 20.5 fm/c

Clustering



Nucleons

